

INFO 579 Project 1 — Requirements Analysis

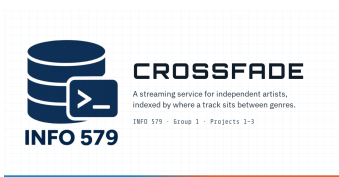
CROSSFADE — a between-genres streaming service for independent artists

Group 1 (info579_group_1) | Herling, Nathan | Pabari, Nidhi Nilesh

Background

The genre scoring models at the centre of CROSSFADE are not hypothetical. They come from **BEARDOWN**, a music genre classifier developed in **INFO 510** (Fall 2025). A convolutional branch reads the log-mel spectrogram of a recording, a multilayer perceptron reads 57 tabular audio features, and a fusion layer (concatenation or gated) joins the two before a softmax over the ten GTZAN genres.

BEARDOWN was re-run and extended in 2026 for the **INFO 698** capstone **FORGE**, a Bayesian posterior observatory that measures whether a trained model's confidence is earned. That work produced five deployed versions that differ in fusion method, embedding width and audio segment length (30 seconds and 3 seconds). Those five versions are the Scoring Model records in this design, and their measured properties (fusion type, embedding dimension, segment length, training corpus and holdout accuracy) supply the example values used throughout this document.

Resource	Link	Social card
CROSSFADE repository and site (INFO 579, this project)	https://github.com/N-Herling-Mk1/INFO_579_Fa26_Proj1_CROSSFADE https://n-herling-mk1.github.io/INFO_579_Fa26_Proj1_CROSSFADE/	 A social card for CROSSFADE. It features a blue database icon with a play button, the text 'CROSSFADE', a subtitle 'A streaming service for independent artists, indexed by where a track sits between genres.', and 'INFO 579' at the bottom.
BEARDOWN repository (INFO 510)	https://github.com/N-Herling-Mk1/INFO_510_Fa25_Final_Proj	 A circular social card for BEARDOWN. It has a red border, a blue background with a yellow paw print, and the text 'B.E.A.R.D.O.W.N.' at the bottom.
FORGE documentation (INFO 698)	https://n-herling-mk1.github.io/INFO_698_documentation/	 A social card for FORGE. It features a circular logo with a flame and the text 'FORGE' below it. To the right, the text reads 'Functional-posterior Observatory for Refinement and Geometric Exploration' and 'Functional-posterior observatory for scientific machine learning'.
FORGE official website	https://forge-observatory.com	 A social card for the FORGE official website. It features a stylized flame logo with the text 'P(θ D):' above it and 'FORGE FUNCTIONAL-POSTERIOR OBSERVATORY' below it.

Data dictionary. Every column in this document, 59 in all, is documented in the project's data dictionary: its data type and size in bytes, the extreme or typical values in `data.xlsx`, why the type is the smallest correct choice, and what the column holds. Online at [the CROSSFADE site, Data dictionary](#); in the repository as `project1/docs/DATA_DICTIONARY.md`. Each entity table in this document links to its section.

Overview

CROSSFADE is a music streaming service for independent **artists** in which the catalogue is indexed by where a **recording** sits between **genres** rather than by a single **genre** label. **Artists** upload their own **recordings** directly and grant the service a streaming licence, which removes the need to negotiate rights with established labels and makes the catalogue consist almost entirely of new and obscure material. Every **track** is passed through a machine learning **scoring model** at the moment of ingest; the model returns a probability distribution across a fixed set of **genres** together with measures of how uncertain and how internally inconsistent that distribution is. **Listeners** do not browse by **genre** name. They create **stations** that describe a **target blend**, for example sixty per cent jazz and forty per cent hip hop, together with a permitted range of ambiguity, and the service returns **recordings** that genuinely occupy that region. Because a **recording** is placed from its audio alone, a **track** uploaded an hour ago is as recommendable as one with a million **plays**, which is the operational advantage the service is built around. The database must therefore manage the **artist** roster and its catalogue, the versioned **scoring models**, the **output each model produced for each track** and the **probability it assigned to each genre**, **listener** accounts and the **stations** they define, and a record of every **play** for the dual purpose of learning **listener** taste and paying **artists** their per stream royalty.

Entities

#	Entity (n=10)	Non-key attributes (n=38)	Attributes
1	Artist	5	Artist ID (PK), Stage Name, Country, Joined Date, Payout Rate, Verified
2	Track	5	Track ID (PK), Artist ID (FK), Title, Duration, Release Date, Audio Checksum, Ingest Status
3	Genre	2	Genre ID (PK), Name, Description
4	Listener	4	Listener ID (PK), Email, Display Name, Plan Tier, Signup Date
5	Scoring Model	7	Model ID (PK), Version Label, Fusion Type, Embedding Dimension, Segment Length, Training Corpus, Holdout Accuracy, Deployed On
6	Station	4	Station ID (PK), Listener ID (FK), Model ID (FK), Name, Minimum Ambiguity, Maximum Ambiguity, Created On
7	Track Score (associative)	5	Track ID (PK, FK), Model ID (PK, FK), Genre ID (FK), Top Probability, Posterior Entropy, Model Divergence, Segment Variance, Scored On
8	Play (associative)	4	Listener ID (PK, FK), Track ID (PK, FK), Played At (PK), Station ID (FK, optional), Seconds Played, Completed, Reaction, Royalty Amount
9	Station Blend (associative)	1	Station ID (PK, FK), Genre ID (PK, FK), Weight
10	Genre Probability (associative)	1	Track ID (PK, FK), Model ID (PK, FK), Genre ID (PK, FK), Probability

The third column counts attributes, not records; it excludes primary key (PK) and foreign key (FK) attributes.

Relationships

- Artist 1:N Track
- Track M:N Scoring Model
- Genre 1:N Track Score
- Listener M:N Track
- Listener 1:N Station
- Scoring Model 1:N Station
- Station 1:N Play
- Station M:N Genre
- Track Score M:N Genre

Entities

Name	Description			
Artist	An independent musician or group that has registered with the service to upload recordings under a direct streaming licence, and is paid a per stream royalty on plays of those recordings. An artist may be registered before uploading a first recording. <i>Types, value ranges and the reason for each type:</i> data dictionary , Artist			
Attributes				
Name	Data Type	Optional	Unique	Primary Key
Artist ID	MEDIUMINT UNSIGNED	No	Yes	Yes
Description				
A unique identifier assigned to each artist when their registration is accepted, used to attribute catalogue and royalty records to that artist.				
Name	Data Type	Optional	Unique	Primary Key
Stage Name	VARCHAR(64)	No	Yes	No
Description				
The public performing name under which the artist releases material and appears throughout the listener-facing service.				
Name	Data Type	Optional	Unique	Primary Key
Country	CHAR(2)	Yes	No	No
Description				
The country in which the artist is based, recorded as its two-letter ISO 3166-1 code and used to determine which royalty payment channels and tax treatments apply.				
Name	Data Type	Optional	Unique	Primary Key
Joined Date	DATE	No	No	No
Description				
The date on which the artist’s registration was accepted and their direct streaming licence took effect.				
Name	Data Type	Optional	Unique	Primary Key
Payout Rate	DECIMAL(5,5)	No	No	No
Description				
The amount in United States dollars the artist is paid for a single qualifying stream, negotiated at registration and fixed for the term of the licence. It is always below one dollar.				
Name	Data Type	Optional	Unique	Primary Key
Verified	BIT(1)	No	No	No
Description				
Whether the service has confirmed the artist’s identity and their right to licence the material they upload.				

Name		Description			
Track		A single audio recording uploaded by an artist, held in the catalogue and made available for streaming once it has been scored by at least one model. The audio file itself is held in an object store under a key computed from the Track ID, so no column stores its location. <i>Types, value ranges and the reason for each type:</i> data dictionary , Track			
Attributes					
Name		Data Type	Optional	Unique	Primary Key
Track ID		INT UNSIGNED	No	Yes	Yes
Description					
A unique identifier assigned to each recording on upload, used to reference that recording across scoring and play records.					
Name		Data Type	Optional	Unique	Primary Key
Artist ID		MEDIUMINT UNSIGNED	No	No	No
Description					
Foreign key referencing Artist. The artist who uploaded the recording and to whom its royalties are paid.					
Name		Data Type	Optional	Unique	Primary Key
Title		VARCHAR(128)	No	No	No
Description					
The name of the recording as the artist wishes it to appear to listeners.					
Name		Data Type	Optional	Unique	Primary Key
Duration		SMALLINT UNSIGNED	No	No	No
Description					
The playing length of the recording in whole seconds, used to decide whether a play qualifies for royalty and to estimate station running time.					
Name		Data Type	Optional	Unique	Primary Key
Release Date		DATE	No	No	No
Description					
The date the artist set for the recording to become available to listeners, which may be later than the date it was uploaded. It takes effect only once the recording has been scored, so a recording awaiting scoring or rejected keeps the date it was given but is not shown to listeners.					
Name		Data Type	Optional	Unique	Primary Key
Audio Checksum		BINARY(32)	No	Yes	No
Description					
The SHA-256 digest of the audio file, used to detect duplicate uploads and to confirm that the stored file has not been altered since it was scored.					
Name		Data Type	Optional	Unique	Primary Key

Ingest Status	ENUM('awaiting scoring', 'scored', 'rejected')	No	No	No
Description				
The stage the recording has reached in the upload pipeline: one of awaiting scoring, scored or rejected.				

Name	Description			
Genre	One of the fixed set of musical categories over which every scoring model produces its probability distribution, forming the coordinate system in which listeners describe the blend they want. <i>Types, value ranges and the reason for each type:</i> data dictionary , Genre			
Attributes				
Name	Data Type	Optional	Unique	Primary Key
Genre ID	TINYINT UNSIGNED	No	Yes	Yes
Description				
A unique identifier for each genre in the fixed set, equal to the class index the scoring models output for that genre, so that a prediction is stored without translation and model output always refers to a controlled category rather than free text. The set is fixed at ten genres, assigned as follows: 0 blues, 1 classical, 2 country, 3 disco, 4 hiphop, 5 jazz, 6 metal, 7 pop, 8 reggae, 9 rock.				
Name	Data Type	Optional	Unique	Primary Key
Name	VARCHAR(9)	No	Yes	No
Description				
The label by which the genre is known to listeners, for example jazz, metal or reggae. The ten names are fixed by the scoring models' output classes; the longest, classical, has nine characters.				
Name	Data Type	Optional	Unique	Primary Key
Description	VARCHAR(80)	Yes	No	No
Description				
A short account of the musical characteristics the category covers, shown to listeners when they are composing a station blend, and kept to one line of 80 characters.				

Name	Description			
Listener	A registered subscriber who streams recordings from the catalogue and defines the stations through which they discover new material. <i>Types, value ranges and the reason for each type:</i> data dictionary , Listener			
Attributes				
Name	Data Type	Optional	Unique	Primary Key
Listener ID	INT UNSIGNED	No	Yes	Yes
Description				
A unique identifier assigned to each subscriber at sign up, used to associate stations and play history with that subscriber.				

Name	Data Type	Optional	Unique	Primary Key
Email	VARCHAR(254)	No	Yes	No
Description				
The address used to sign in, to recover the account and to send billing correspondence.				
Name	Data Type	Optional	Unique	Primary Key
Display Name	VARCHAR(32)	No	No	No
Description				
The name the subscriber chooses to be shown by when a station they created is shared with other subscribers.				
Name	Data Type	Optional	Unique	Primary Key
Plan Tier	ENUM('free', 'standard', 'premium')	No	No	No
Description				
The subscription level held by the listener, one of free, standard or premium, which determines the monthly charge and the limits placed on station creation and offline listening.				
Name	Data Type	Optional	Unique	Primary Key
Signup Date	DATE	No	No	No
Description				
The date on which the subscriber first registered, used to anchor the monthly billing cycle.				

Name	Description			
Scoring Model	A specific trained and deployed version of the machine learning system that places recordings in genre space. Several versions are in service at once because they differ in architecture and therefore disagree with one another, and that disagreement is itself used as a signal. <i>Types, value ranges and the reason for each type:</i> data dictionary , Scoring Model			
Attributes				
Name	Data Type	Optional	Unique	Primary Key
Model ID	TINYINT UNSIGNED	No	Yes	Yes
Description				
A unique identifier for each deployed model version, assigned in order of deployment starting at one, carried on every score so that no result is ever separated from the model that produced it.				
Name	Data Type	Optional	Unique	Primary Key
Version Label	VARCHAR(32)	No	Yes	No
Description				
The human readable release name of the model version, used in internal reporting and in the artist analytics product.				
Name	Data Type	Optional	Unique	Primary Key
Fusion Type	ENUM('concat', 'gated')	No	No	No

Description				
The method by which the model combines its audio branch with its tabular branch, either concatenation (concat) or gated fusion (gated), which is one of the principal architectural differences between versions.				
Name	Data Type	Optional	Unique	Primary Key
Embedding Dimension	SMALLINT UNSIGNED	No	No	No
Description				
The width of the representation the model produces for a recording, which determines the geometry of the space in which similarity is measured.				
Name	Data Type	Optional	Unique	Primary Key
Segment Length	TINYINT UNSIGNED	No	No	No
Description				
The length in whole seconds of the audio window the model scores at a time; shorter windows allow a recording to be shown to shift in character across its own duration.				
Name	Data Type	Optional	Unique	Primary Key
Training Corpus	VARCHAR(16)	No	No	No
Description				
The name of the labelled collection the model version was trained on, recorded so that results from differently trained versions are never silently compared. Corpus names are short identifiers such as GTZAN; 16 characters covers the common public sets.				
Name	Data Type	Optional	Unique	Primary Key
Holdout Accuracy	DECIMAL(5,4)	No	No	No
Description				
The proportion of held out recordings the model version classified correctly, used to decide how much weight its output carries in a blended placement.				
Name	Data Type	Optional	Unique	Primary Key
Deployed On	DATE	No	No	No
Description				
The date the model version was placed in service and began scoring incoming uploads.				

Name	Description			
Station	A standing description of a region of genre space created by a listener. A station stores the blend and the ambiguity range the listener wants rather than a fixed list of recordings, so its contents change as the catalogue grows. <i>Types, value ranges and the reason for each type:</i> data dictionary , Station			
Attributes				
Name	Data Type	Optional	Unique	Primary Key
Station ID	INT UNSIGNED	No	Yes	Yes
Description				

A unique identifier for each station, used to record which station was responsible for surfacing a given play.

Name	Data Type	Optional	Unique	Primary Key
Listener ID	INT UNSIGNED	No	No	No

Description

Foreign key referencing Listener. The subscriber who created the station and to whose library it belongs.

Name	Data Type	Optional	Unique	Primary Key
Model ID	TINYINT UNSIGNED	No	No	No

Description

Foreign key referencing Scoring Model. The model version whose output the station is evaluated against.

Name	Data Type	Optional	Unique	Primary Key
Name	VARCHAR(64)	No	No	No

Description

The name the listener gives the station, which appears in their library and in any share link.

Name	Data Type	Optional	Unique	Primary Key
Minimum Ambiguity	DECIMAL(3,2)	No	No	No

Description

The lower bound, on the zero to one posterior entropy scale, on how unsettled a recording's placement must be to qualify; raising it excludes recordings that sit squarely inside a single genre. It may not exceed Maximum Ambiguity.

Name	Data Type	Optional	Unique	Primary Key
Maximum Ambiguity	DECIMAL(3,2)	No	No	No

Description

The upper bound, on the zero to one posterior entropy scale, on how unsettled a recording's placement may be; lowering it excludes recordings the models cannot place with any confidence.

Name	Data Type	Optional	Unique	Primary Key
Created On	DATE	No	No	No

Description

The date the listener created the station, used to order their library and to measure how long a station remains in use.

Associative entities

The primary key of each associative entity is formed from its foreign keys, the primary keys of the entities it connects, and no associative entity has a surrogate key. Track Score, Station Blend and Genre Probability follow that rule exactly, so each participates in two identifying relationships. Genre Probability connects Track Score to Genre, so its key is the two-part key of Track Score followed by Genre ID. Each key component is marked Unique = No, because a single component repeats freely; only the combination is unique. Two of the associative entities also carry a third relationship, which gives the model its two ternary structures: Track, Scoring Model and Genre meet at Track Score, and Listener, Track and Station meet at Play. In both cases the third foreign key stays outside the primary key. Genre ID is excluded because the top genre is determined by the pair of recording and model, so adding it to the key would permit two top genres for one pair. Station ID is excluded because it is optional, and a primary key column can never be empty.

Play follows the rule and adds exactly one column. Its foreign keys, Listener ID and Track ID, are both in its key, but together they cannot identify a play, because a listener streams the same recording many times. A surrogate Play ID would solve that only by abandoning the rule, which is what the feedback on the first draft corrected. Played At, the moment the stream began, completes the key instead. The course's own Unit 2 example uses the same pattern: in its student grades table, student 12345678 takes Computing Mathematics in 2019 and again in 2020, so a grade is identified by student and course only together with Year and Sem. The key is unique under two stated rules. Played At is recorded in UTC, so a daylight-saving change can never repeat a time. A second event with the same listener, recording and start time is a duplicate report of the same play, and the key rejects it, which is what keeps a double-counted royalty out of the ledger.

Name	Description			
Track Score	The result one scoring model produced for one recording. This is the associative entity that resolves the many to many relationship between recordings and scoring models, and it is the parent record for every model output: the full genre distribution is held in Genre Probability rows that belong to it. No genre is ever recorded against a recording on its own, because a placement is a property of the pair of recording and model and not of the recording alone. The full distribution is held in Genre Probability; Genre ID, Top Probability, Posterior Entropy and Model Divergence are summaries derived from it and stored on this row so that stations can filter without reading ten rows per score. A result is written once, in the same transaction as its distribution, and is replaced whole on a rescore, so the stored summaries cannot drift from the distribution. The model's vector representation of the recording is held in a vector store under a key computed from the same Track ID and Model ID, so no column stores it. <i>Types, value ranges and the reason for each type:</i> data dictionary , Track Score			
Attributes				
Name	Data Type	Optional	Unique	Primary Key
Track ID	INT UNSIGNED	No	No	Yes
Description				
Part of the composite primary key; foreign key referencing Track. The recording that was scored.				
Name	Data Type	Optional	Unique	Primary Key
Model ID	TINYINT UNSIGNED	No	No	Yes
Description				

Part of the composite primary key; foreign key referencing Scoring Model. The model version that produced the result. Together with Track ID it allows at most one result per recording and model; rescoring with the same version replaces that result, and a retrained model is deployed as a new version with its own Model ID.

Name	Data Type	Optional	Unique	Primary Key
Genre ID	TINYINT UNSIGNED	No	No	No
Description				
Foreign key referencing Genre, not part of the primary key. The genre the model judged most likely for this recording; derived as the genre with the highest probability in Genre Probability for the same recording and model.				
Name	Data Type	Optional	Unique	Primary Key
Top Probability	DECIMAL(5,4)	No	No	No
Description				
The probability the model assigned to the genre it judged most likely, which is how confident that single placement is; derived as the largest probability in Genre Probability for the same recording and model.				
Name	Data Type	Optional	Unique	Primary Key
Posterior Entropy	DECIMAL(5,4)	No	No	No
Description				
How spread out the model's probability distribution is across all genres, expressed as the Shannon entropy divided by its maximum possible value so that it lies between zero and one; a high value means the recording sits between categories rather than inside one. Derived from the distribution in Genre Probability.				
Name	Data Type	Optional	Unique	Primary Key
Model Divergence	DECIMAL(5,4)	Yes	No	No
Description				
How far this model's distribution sits from those produced by the other model versions that have scored the same recording. It is the mean, over every other result for that recording, of the Jensen-Shannon divergence in base two between the two distributions; each divergence lies between zero and one, so the mean does too. It is empty while only one model has scored the recording, because there is nothing to compare against, and zero would wrongly claim perfect agreement. This is the cross model disagreement signal referred to in the Scoring Model definition. Derived from the distributions in Genre Probability; because it depends on the other models' results for the same recording, it is recomputed for every result of that recording whenever any model scores it.				
Name	Data Type	Optional	Unique	Primary Key
Segment Variance	DECIMAL(4,4)	Yes	No	No
Description				
How much the placement moves between one audio window and the next within the same recording, measured as the variance of the top genre probability across windows and therefore never above one quarter, which identifies tracks that change character partway through.				
Name	Data Type	Optional	Unique	Primary Key
Scored On	DATETIME	No	No	No

Description
The moment the scoring run completed, used to establish which results predate a model redeployment and must be recomputed.

Name		Description		
Play		A single occasion on which a listener streamed a recording. This is the associative entity that resolves the many to many relationship between listeners and recordings, and it serves two purposes at once: it is the record from which listener taste is inferred and it is the billable event on which the artist royalty is calculated. <i>Types, value ranges and the reason for each type:</i> data dictionary , Play		
Attributes				
Name	Data Type	Optional	Unique	Primary Key
Listener ID	INT UNSIGNED	No	No	Yes
Description				
Part of the composite primary key; foreign key referencing Listener. The subscriber who streamed the recording.				
Name	Data Type	Optional	Unique	Primary Key
Track ID	INT UNSIGNED	No	No	Yes
Description				
Part of the composite primary key; foreign key referencing Track. The recording that was streamed.				
Name	Data Type	Optional	Unique	Primary Key
Played At	DATETIME	No	No	Yes
Description				
Part of the composite primary key. The moment the stream began, recorded in UTC so that a daylight-saving change never repeats a value, and used to order listening history and to assign the play to a billing period. A listener may stream the same recording many times, so Listener ID and Track ID alone do not identify a play; the start time completes the key, and a repeated event with the same three values is rejected as a duplicate when the royalty ledger is reconciled.				
Name	Data Type	Optional	Unique	Primary Key
Station ID	INT UNSIGNED	Yes	No	No
Description				
Foreign key referencing Station, not part of the primary key. The station that surfaced the recording, left empty when the listener chose it directly. Because stations can be shared, the station may belong to a different listener from the one recorded on the play.				
Name	Data Type	Optional	Unique	Primary Key
Seconds Played	SMALLINT UNSIGNED	No	No	No
Description				
How much of the recording was actually streamed, which determines both whether the play qualifies for royalty and how strong a taste signal it carries.				

Name	Data Type	Optional	Unique	Primary Key
Completed	BIT(1)	No	No	No
Description				
Whether the listener heard the recording to its end, distinguishing a genuine listen from an abandoned one. Derived: true exactly when Seconds Played equals the Duration of the recording. It is stored so that listening history can be filtered to finished plays without a join to Track, and it is set only by the procedure that records the play (I7).				
Name	Data Type	Optional	Unique	Primary Key
Reaction	ENUM('saved', 'skipped')	Yes	No	No
Description				
An explicit response from the listener where one was given, either saved or skipped, which is treated as a stronger signal than duration alone.				
Name	Data Type	Optional	Unique	Primary Key
Royalty Amount	DECIMAL(5,5)	No	No	No
Description				
The amount in United States dollars accrued to the artist for this play. A snapshot rather than a derived value: it is the artist's Payout Rate as it stood at the time of a qualifying play of 30 seconds or more, and zero otherwise, copied so that a later rate change never rewrites the royalty ledger (I7).				

Name	Description			
Station Blend	The weight one station gives to one genre. This is the associative entity that resolves the many to many relationship between stations and genres: a station blends several genres and a genre appears in many stations. Holding one weight per row keeps every value atomic, which a single multi-valued blend field on the station could not. <i>Types, value ranges and the reason for each type:</i> data dictionary , Station Blend			
Attributes				
Name	Data Type	Optional	Unique	Primary Key
Station ID	INT UNSIGNED	No	No	Yes
Description				
Part of the composite primary key; foreign key referencing Station. The station whose blend this row belongs to.				
Name	Data Type	Optional	Unique	Primary Key
Genre ID	TINYINT UNSIGNED	No	No	Yes
Description				
Part of the composite primary key; foreign key referencing Genre. The genre being weighted. Together with Station ID it allows at most one weight per genre in a station, so a single weight can be adjusted without rewriting the rest of the blend.				
Name	Data Type	Optional	Unique	Primary Key

Weight	DECIMAL(3,2)	No	No	No
Description				
The share of the station's blend given to this genre, greater than zero and at most one. The weights belonging to one station sum to exactly 1.00; because this is a rule across rows, it is enforced when the blend is written (see Integrity rules), not by a single-row check.				

Name		Description		
Genre Probability		The probability one scoring model assigned to one genre for one recording. This is the associative entity that resolves the many to many relationship between scoring results and genres: a result assigns a probability to every genre, and a genre receives a probability in every result. It holds the full distribution that stations match against, which the single top genre on Track Score cannot express. <i>Types, value ranges and the reason for each type:</i> data dictionary , Genre Probability		
Attributes				
Name	Data Type	Optional	Unique	Primary Key
Track ID	INT UNSIGNED	No	No	Yes
Description				
Part of the composite primary key. Together with Model ID it forms a composite foreign key referencing Track Score, so a probability can only be recorded for a recording and model pair that has a scoring result.				
Name	Data Type	Optional	Unique	Primary Key
Model ID	TINYINT UNSIGNED	No	No	Yes
Description				
Part of the composite primary key and of the composite foreign key referencing Track Score. The model version that produced the distribution.				
Name	Data Type	Optional	Unique	Primary Key
Genre ID	TINYINT UNSIGNED	No	No	Yes
Description				
Part of the composite primary key; foreign key referencing Genre. The genre this probability belongs to.				
Name	Data Type	Optional	Unique	Primary Key
Probability	DECIMAL(5,4)	No	No	No
Description				
The probability, between zero and one, that the model assigned to this genre for this recording. The ten probabilities of one result sum to one, within rounding.				

Relationships

Parent	Child	Cardinality	Description
Artist	Track	1:N	An artist contributes zero or more recordings to the catalogue (none between registration and the first upload), and every recording belongs to exactly one artist, who is the party the royalty is paid to.
Track	Scoring Model	M:N	A recording is scored by zero or more model versions (none while it is awaiting scoring, at least one before it is released) and a model version scores many recordings. The relationship is resolved by the associative entity Track Score, whose primary key is Track ID and Model ID.
Genre	Track Score	1:N	Each scoring result names the single genre the model judged most likely, and one genre is named by many scoring results. The genre is recorded on the result, as a foreign key outside the primary key, and never on the recording itself.
Listener	Track	M:N	A listener streams many recordings and a recording is streamed by many listeners. The relationship is resolved by the associative entity Play, whose primary key is Listener ID, Track ID and Played At.
Listener	Station	1:N	A listener creates zero or more stations and every station belongs to exactly one listener.
Scoring Model	Station	1:N	Each station is evaluated against the output of one nominated model version, so that a station's contents do not change meaning when a different version scores the catalogue. One model version serves many stations.
Station	Play	1:N	A station is responsible for many streaming events, while a play may have arisen from a station or from the listener choosing a recording directly, in which case Station ID on the play is left empty. A station may be shared, so it can surface plays for listeners other than its creator.
Station	Genre	M:N	A station blends one or more genres and a genre appears in many stations. The relationship is resolved by the associative entity Station Blend, whose primary key is Station ID and Genre ID.

Parent	Child	Cardinality	Description
Track Score	Genre	M:N	A scoring result assigns a probability to every genre and a genre receives a probability in every result. The relationship is resolved by the associative entity Genre Probability, whose primary key is Track ID, Model ID and Genre ID.

Integrity rules

Keys and single-row checks cannot express every rule. The rules below span rows or tables. Each is enforced by a stored procedure that writes all the affected rows in one transaction and either computes the dependent values itself or validates them before it commits. A transaction alone guarantees only that the rows are written together, not that they agree, so the procedures do not accept derived values as input. The application account is granted EXECUTE on these procedures and no direct INSERT, UPDATE or DELETE on Track Score, Genre Probability or Station Blend, so the procedures are the only write path to those tables.

#	Rule	Enforcement
11	A station has at least one blend row, and its weights sum to exactly 1.00.	The procedure that saves a blend replaces all of the station's rows at once and rolls back unless there is at least one row and $\text{SUM}(\text{Weight}) = 1.00$. A station and its first blend are created in the same call. Weights are held in hundredths, so a three-genre blend is stored as 0.34, 0.33 and 0.33, never as three equal thirds.
12	A scoring result has exactly one probability for each of the ten genres, and they sum to one.	The scoring procedure takes the ten probabilities as its input and rolls back unless there are exactly as many rows as there are Genre records, one per genre, summing to one within 0.0010, the rounding allowance for ten values held to four decimal places.
13	Genre ID, Top Probability, Posterior Entropy and Model Divergence on Track Score agree with the probabilities in Genre Probability.	The scoring procedure computes all four from the probability rows it has just written rather than accepting them as input: the most likely genre, its probability, the normalised Shannon entropy, and the mean base-two Jensen–Shannon divergence against every other result for the same recording, left empty when there is no other result. Because divergence depends on those other results, the procedure recomputes it for all of the recording's results. A rescore replaces the result and its distribution together.
14	Minimum Ambiguity does not exceed Maximum Ambiguity.	A single-row CHECK constraint on Station.
15	A recording's Ingest Status is scored only if it has at least one Track Score.	Set in the scoring transaction that inserts the first result.

#	Rule	Enforcement
I6	A play's Station ID may refer to a station created by another listener.	Deliberately not constrained by the database: stations can be shared, so the station's Listener ID need not match the play's. Whether a listener may use a given station is a sharing permission checked by the application.
I7	Completed is true exactly when Seconds Played equals the recording's Duration, and Royalty Amount equals the artist's Payout Rate for a play of 30 seconds or more and zero otherwise.	The procedure that records a play computes both from the Track and Artist rows rather than accepting them as input, as I3 does for Track Score. Royalty Amount is computed once, when the play is recorded, and never updated.

Index and Colour Coding

Every entity, associative entity and relationship in the requirements analysis, colour coded by kind. The same colour marks a name wherever it appears in these tables.

■ Entity ■ Associative entity ■ Relationship

Table 1 — Entities (6)					
#	Name	Attributes	Primary Key	Key Type	Example
E1	Artist	5	Artist ID	MEDIUMINT UNSIGNED	1042 (Velvet Static)
E2	Track	5	Track ID	INT UNSIGNED	4817 (Low Tide Signal)
E3	Genre	2	Genre ID	TINYINT UNSIGNED	5 (jazz)
E4	Listener	4	Listener ID	INT UNSIGNED	20931 (jrivers)
E5	Scoring Model	7	Model ID	TINYINT UNSIGNED	5 (beard-down_3sec)
E6	Station	4	Station ID	INT UNSIGNED	7710 (Late Jazz Hop)

Genre ID and Model ID are the only keys whose values are fixed by the design rather than generated. Each value below is one record of the Genre or Scoring Model table; child tables store only the number, and a join on the key returns the name. Genre ID equals the class index the scoring models output. Model ID is assigned in order of deployment.

Table 2 — Key reference for controlled keys			
Key	Value	Name	Source
Genre ID	0	blues	model class index
Genre ID	1	classical	model class index
Genre ID	2	country	model class index
Genre ID	3	disco	model class index
Genre ID	4	hiphop	model class index
Genre ID	5	jazz	model class index
Genre ID	6	metal	model class index
Genre ID	7	pop	model class index
Genre ID	8	reggae	model class index
Genre ID	9	rock	model class index
Scoring Model ID	1	beardown	deployment order
Scoring Model ID	2	beardown_rrm_cfg12	deployment order
Scoring Model ID	3	beardown_rrm_cfg17	deployment order
Scoring Model ID	4	beardown_rrm	deployment order
Scoring Model ID	5	beardown_3sec	deployment order

Table 3 — Associative entities (4)

#	Name	Attributes	Primary Key	Resolves
A1	Track Score	5	Track ID + Model ID	Track M:N Scoring Model (R2)
A2	Play	4	Listener ID + Track ID + Played At	Listener M:N Track (R4)
A3	Station Blend	1	Station ID + Genre ID	Station M:N Genre (R8)
A4	Genre Probability	1	Track ID + Model ID + Genre ID	Track Score M:N Genre (R9)

Table 4 — Relationships (9)

#	Parent	Child	Cardinality	Resolved by
R1	Artist	Track	1:N	—
R2	Track	Scoring Model	M:N	Track Score
R3	Genre	Track Score	1:N	—
R4	Listener	Track	M:N	Play
R5	Listener	Station	1:N	—
R6	Scoring Model	Station	1:N	—
R7	Station	Play	1:N	—
R8	Station	Genre	M:N	Station Blend
R9	Track Score	Genre	M:N	Genre Probability

Table 5 — Resolution of each many-to-many relationship

Relationship	Before	Via	After
R2	Track M:N Scoring Model	Track Score	Track 1:N Track Score Scoring Model 1:N Track Score
R4	Listener M:N Track	Play	Listener 1:N Play Track 1:N Play
R8	Station M:N Genre	Station Blend	Station 1:N Station Blend Genre 1:N Station Blend
R9	Track Score M:N Genre	Genre Probability	Track Score 1:N Genre Probability Genre 1:N Genre Probability

Every foreign key in the design, with the relationship it implements. A foreign key that is part of its owner's primary key implements an identifying relationship; one that is not implements a non-identifying relationship.

Table 6 — Foreign keys (14 columns)					
Owner	Foreign key	References	In PK	Rel.	Relationship type
Track	Artist ID	Artist	No	R1	non-identifying
Station	Listener ID	Listener	No	R5	non-identifying
Station	Model ID	Scoring Model	No	R6	non-identifying
Track Score	Track ID	Track	Yes	R2	identifying
Track Score	Model ID	Scoring Model	Yes	R2	identifying
Track Score	Genre ID	Genre	No	R3	non-identifying
Play	Listener ID	Listener	Yes	R4	identifying
Play	Track ID	Track	Yes	R4	identifying
Play	Station ID	Station	No	R7	non-identifying, optional
Station Blend	Station ID	Station	Yes	R8	identifying
Station Blend	Genre ID	Genre	Yes	R8	identifying
Genre Probability	Track ID + Model ID	Track Score	Yes	R9	identifying, composite
Genre Probability	Genre ID	Genre	Yes	R9	identifying

Document Inventory

Everything this document currently contains, followed by the evidence that the design meets each stated design requirement, a full attribute inventory and the data types used.

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Background, with BEARDOWN and FORGE links	1
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Table 7 — Design requirement checks				
Requirement	Required	This design	Status	Evidence
Non-associative entities	5 to 7	6	Met	Table 1
Attributes, excluding primary and foreign keys	≥ 20	38	Met	Table 8
Records in the non-associative entities, at least 5 in each	≥ 50 , ≥ 5 each	54, fewest 5	Met	data.xlsx
Many-to-many relationships resolved with an associative entity	≥ 1	4	Met	Tables 3, 4, 5, 6

Requirement 1. The design has 6 non-associative entities (Table 1). The 4 associative entities are listed separately (Table 3) and are not counted.

Requirement 2. The design has 38 attributes excluding primary and foreign keys: 27 on the entities and 11 on the associative entities. Primary keys (Tables 1 and 3) and the 14 foreign key columns (Table 6) are not counted. Played At is counted as part of Play's primary key and is therefore excluded as well, although it is not a foreign key.

Requirement 3. data.xlsx holds 54 records across the six non-associative entities: Artist 8, Track 15, Genre 10, Listener 8, Scoring Model 5 and Station 8, so every entity has at least 5. The four associative entities add a further 347 rows (Track Score 27, Genre Probability 270, Station Blend 19, Play 31), which are not counted.

Requirement 4. Four of the nine relationships are many-to-many (Table 4). Each is resolved by an associative entity (Table 3) that turns it into two one-to-many identifying relationships (Table 5), with the primary key of the associative entity composed of the primary keys of the two entities it connects (Table 6).

Table 8 lists every attribute on its own row with its data type, the range that type admits and an example value. Scoring Model values are those of a real deployed version, the 3-second segment model trained on GTZAN; the remaining values are illustrative, with the Track Score figures computed from one example posterior so that they agree with one another and fall inside the example station's ambiguity range.

#	Owner	Attribute	Data Type	Range	Example
1	Artist	Stage Name	VARCHAR(64)	up to 64 characters	Velvet Static
2	Artist	Country	CHAR(2)	ISO 3166-1 alpha-2 code	US
3	Artist	Joined Date	DATE	calendar date	2026-03-14
4	Artist	Payout Rate	DECIMAL(5,5)	0.00000 to 0.99999 USD	0.00400
5	Artist	Verified	BIT(1)	0 or 1	1
6	Track	Title	VARCHAR(128)	up to 128 characters	Low Tide Signal
7	Track	Duration	SMALLINT UNSIGNED	0 to 65,535 s	214
8	Track	Release Date	DATE	calendar date	2026-04-02
9	Track	Audio Checksum	BINARY(32)	32-byte SHA-256 digest	0x9F3A...C21E
10	Track	Ingest Status	ENUM('awaiting scoring', 'scored', 'rejected')	one of 3 values	'scored'
11	Genre	Name	VARCHAR(9)	one of 10 names, longest 'classical' (9)	jazz
12	Genre	Description	VARCHAR(80)	up to 80 characters	Swing, improvisation and extended harmony
13	Listener	Email	VARCHAR(254)	up to 254 characters	j.rivera@example.com
14	Listener	Display Name	VARCHAR(32)	up to 32 characters	jrivers
15	Listener	Plan Tier	ENUM('free', 'standard', 'premium')	one of 3 values	'standard'
16	Listener	Signup Date	DATE	calendar date	2026-05-09
17	Scoring Model	Version Label	VARCHAR(32)	up to 32 characters	beardown_3sec
18	Scoring Model	Fusion Type	ENUM('concat', 'gated')	one of 2 values	'concat'

#	Owner	Attribute	Data Type	Range	Example
19	Scoring Model	Embedding Dimension	SMALLINT UNSIGNED	0 to 65,535	384
20	Scoring Model	Segment Length	TINYINT UNSIGNED	0 to 255 s	3
21	Scoring Model	Training Corpus	VARCHAR(16)	up to 16 characters	GTZAN
22	Scoring Model	Holdout Accuracy	DECIMAL(5,4)	0.0000 to 1.0000	0.7887
23	Scoring Model	Deployed On	DATE	calendar date	2026-06-23
24	Station	Name	VARCHAR(64)	up to 64 characters	Late Jazz Hop
25	Station	Minimum Ambiguity	DECIMAL(3,2)	0.00 to 1.00	0.35
26	Station	Maximum Ambiguity	DECIMAL(3,2)	0.00 to 1.00	0.80
27	Station	Created On	DATE	calendar date	2026-08-01
28	Track Score	Top Probability	DECIMAL(5,4)	0.0000 to 1.0000	0.5800
29	Track Score	Posterior Entropy	DECIMAL(5,4)	0.0000 to 1.0000	0.4619
30	Track Score	Model Divergence	DECIMAL(5,4)	0.0000 to 1.0000, or empty	0.0198
31	Track Score	Segment Variance	DECIMAL(4,4)	0.0000 to 0.2500	0.0051
32	Track Score	Scored On	DATETIME	date and time	2026-07-01 09:14:52
33	Play	Seconds Played	SMALLINT UNSIGNED	0 to 65,535 s	214
34	Play	Completed	BIT(1)	0 or 1	1
35	Play	Reaction	ENUM('saved', 'skipped')	one of 2 values	'saved'
36	Play	Royalty Amount	DECIMAL(5,5)	0.00000 to 0.99999 USD	0.00400
37	Station Blend	Weight	DECIMAL(3,2)	0.01 to 1.00	0.60
38	Genre Probability	Probability	DECIMAL(5,4)	0.0000 to 1.0000	0.5800

Every data type in this document is taken from the MySQL type set presented in the Unit 4 slides, and each column is given the smallest type whose range covers every value the attribute can legitimately hold. Integer keys are declared unsigned because no identifier is negative. Strings follow four rules. A value whose length never varies is CHAR (Country) or BINARY (Audio Checksum), which saves VARCHAR's length byte. A value drawn from a closed set gets a VARCHAR sized to its longest member: Genre Name is one of ten fixed names, the longest nine. A value that can be computed from other columns is not stored at all: the object-store key of a recording's audio and the vector-store key of a model's representation are both computed from the primary key, so neither has a column. Free text entered by people (names, titles, email) gets a ceiling set by policy or standard, since VARCHAR stores only the characters used plus a length byte. Ranges that a type cannot enforce on its own, such as a probability lying between zero and one, are stated in the attribute description and are declared as 16 CHECK constraints in the schema script from which the physical model was generated (crossfade_schema.sql). The physical model diagram shows the NOT NULL, UNIQUE, primary key and foreign key constraints; MySQL Workbench does not display CHECK constraints or the UNSIGNED attribute in a diagram. Two types from the slides are deliberately not used: TIMESTAMP, whose range ends in January 2038, well within the life of a royalty ledger, so DATETIME is used for moments in time. That is the one place parsimony is knowingly traded: the slides give DATETIME 8 bytes against TIMESTAMP's 4 (MySQL 5.6.4 and later stores DATETIME in 5), and Played At is part of Play's key, but a play recorded after 2038 must still be payable. FLOAT and DOUBLE are not used either, because every fractional value here has a known precision and DECIMAL stores it exactly.

Table 9 — Data types used, by family			
Family	Types used	Columns	Slide source
Integer	INT UNSIGNED, MEDIUMINT UNSIGNED, SMALLINT UNSIGNED, TINYINT UNSIGNED	24	Unit 4, Numeric Data Types
Fixed-point	DECIMAL(3,2), DECIMAL(4,4), DECIMAL(5,4), DECIMAL(5,5)	11	Unit 4, Numeric Data Types
String	CHAR(2), VARCHAR(9), VARCHAR(16), VARCHAR(32), VARCHAR(64), VARCHAR(80), VARCHAR(128), VARCHAR(254)	10	Unit 4, String Data Types
Binary	BINARY(32)	1	Unit 4, Binary Data Types
Date and time	DATE, DATETIME	7	Unit 4, Date and Time Types
Enumeration	ENUM (4 definitions)	4	Unit 4, Other Data Types
Boolean	BIT(1)	2	Unit 4, Other Data Types